LAB 8: POINTER

OBJECTIVE

-> To be familier with pointer memory management concept in C.

Program # 1

PROBLEM ANALYSIS:

To write a program to swap two numbers using user defined function and using call by reference approach we use correct concept and way of pointer.

ALGORITHM:

1. Start
2. Declare two variables
3. Read values
4. Call swap function by passing address of a and b.
5. In function assign and store.
6. Return to main function
7. Declare swapped value
8. STOP

C-PROGRAM

```
# include <stdio.h>
void swap by Ref (int *p, int *q)
{int temp;
 *p = *q
 *q = temp;
}
int main ( )
{
int a = 10, b = 20
printf ("Before swap a = %d & b = %d \n", a, b);
swap by Ref (&a, &b);
printf ("After swap: a = %d & b = %d \n", a, b);
return 0;
}
```

OUTPUT:

Before swap: a = 10 & b = 20
After swap: a = 20 & b = 10

RESULT

The values of a & b were swapped.

PROGRAM # 2

PROBLEM ANALYSIS

To write a program in C to illustrate pointer arithemetic we aims to demonstrate how pointers can be increment or decrement to access different memory locations.

ALGORITHM

1. START
2. Declare array & initialize variable
3. Declare pointer and assign in to array based address
4. Display initial address and value.
5. Increment pointer cp = p+1
6. Display new address and value
7. STOP.

C-PROGRAM:

```
# include <stdio.h>
int main ( )
{
float *fp, b=89.6;
printf("fp is pointing to: %v\n", fp);
fp++;
printf("fp++ is pointing to %v\n", fp);
fp = fp+2;
printf("fp+2 is pointing to %v\n", fp);
return 0;
}
```

OUTPUT

fp is pointing to : 6422084
fp++ is pointing to : 6422088
fp+2 is pointing to : 6422096

RESULT

fp, fp++, f+2 were displayed successfully and p.arithmetic was demonstrated.

PROGRAM # 3

PROBLEM ANALYSIS

To write a program in C to find length of string using user defined function and pointer to calculate length from user defined function and pointer to display from main ( ) function, we need to emphasize pointer reversal and returning the result to main function.

ALGORITHM

1. START
2. Declarea string
3. Read string input
4. Call function by passing pointer
5. In function (Initialize Encounter to 0, Tranver string and Increment counter)
6. Return length to main function
7. Display length
8. STOP

C-PROGRAM

```
# include <stdio.h >
int Gret string length (char *p )
{
int. length = 0;
for (int i=0; *(p+i)? = '\0'; i++ )
{
length ++;
}
Return length;
}
int main ( )
{
```

```
char S[200]
put S("Enter string:");
get s(s);
len = net string length(s);
printf("length of string = %d\n", len);
return 0;
}
```

OUTPUT:

Enter string = Meghan
length of string = 6

RESULT

The length of string was counted successfully

PROGRAM#4

PROBLEM ANALYSIS

To write a program in a short an array in ascending order using user defined function and pointer and to display the result from main( ) function we need to use the pointer along with array shorting methods.

ALGORITHM

1. START
2. Declare array & input elements
3. Call shorting function
4. In funcing (use nested loop, compare and swap in order)
5. Return to main function
6. Display shorted Array
7. STOP

C-PROGRAM

```
# include <stdio.h>
void short Array (int arr, int n)
{
int i, j, temp;
for (i=0; i<n-1; i++)
{
for (j = 1; j<n; i++)
{
if (*(arr+i) * (arr+j))
{
temp = *(arr+j);
*(arr+i) = *(arr+j);
*(arr+j) = temp;
```

```
int main ( )
{
int arr [100], n,j);
printf("Enter numbers of elements: ");
Scanf("%d", &n);
printf("Enter %d elements:\n", n);
for (i=0; i<n; i++)
{
Scanf("%d", (arr+i));
}
Short array (arr, n)
printf("shorted Array is:\n");
for (i=0; i<n; i++)
{
printf("%d, *(arr+i));
}return 0;
}
```

OUTPUT

Enter numbers of elements : 4
Enter 4 elements:
1 2 0 8

Shorted array is
0 1 2 8

RESULT

The array elements where shorted successfully.

CONCLUSION

↳ In this Lab session we learned to use structure in C-program along with pointers and user-defined function successfully.

CamScanner